



Probability

chance of occurring some event

Basic Terms

① Random Experiment

→ Results or outcomes cannot be predicted in advance

→ It has multiple outcomes or results.

Examples:

① Tossing a fair coin → Random Experiment

Possible outcomes: Heads, Tail

② Tossing a fair coin twice → Random Experiment

∴ possible outcomes : { HH, HT, TH, TT }

② Sample Space

Set of all possible outcomes or results.

S or Ω .

It is denoted by

Example: ① Tossing a coin → Random Experiment

∴ Sample Space,

$$S = \{ H, T \}$$

② Rolling a die → Random experiment.

∴ Sample space, $S = \{ 1, 2, 3, 4, 5, 6 \}$

③ Trial and Event

Trial: Single performance of a random experiment

Event: Result of ~~trial~~ or outcome of a trial or a random experiment. It is also known as subset of Sample space.

→ Simple Event
→ Composite event.

→ Event is denoted by capital Alphabets
→ Events are defined on S .

E.g. Rolling a die → Random experiment

$$S = \{1, 2, 3, 4, 5, 6\}$$

Let, $A =$ occurrence of 1 → $\{1\}$ ⇒ simple event
or results

$B =$ " " even no. → $\{2, 4, 6\}$
↓
Composite event.

④ Equally likely events:
Two or more events are ~~same~~ said to be equally likely events if they have equal chance of occurrence

① e.g. Tossing a coin

$$S = \{H, T\}$$

Let, $A =$ occurrence of head: $\{H\}$

$B =$ occurrence of tail: $\{T\}$

Here, A and B are equally likely events.

② Rolling a die

$$S = \{1, 2, 3, 4, 5, 6\}$$

Let, A = Occurrence of 1

B = Occurrence of 2

C = Occurrence of odd no.

D = Occurrence of even no.

Here, A and B are equally likely events.

Also, C and D are equally likely. But

A and C ~~or B and D~~ are not equally likely.

④ Mutually Exclusive Events:

Two or more events are said to be mutually exclusive events if they do not occur at a same time in a same random experiment.

Eg; Rolling a die;

$S = \{1, 2, 3, 4, 5, 6\}$

Let, A = Occurrence of 1

B = Occurrence of 2

C = Occurrence of odd no.

D = Occurrence of even no.

Here, A and B are mutually exclusive.

C and D are mutually exclusive.

but A and C are not mutually exclusive as 1 can be in C as well.

(vi) ~~Exhausted~~ Exhaustive Cases or Events: (n)

Total no. of ~~case~~ possible outcomes or cases or events of a random experiment.

e.g. Cardinal no. of Sample space (S),
i.e., $n(S)$.

(vii) Favorable Cases or Events: (m)
No. of cases or events that are favorable to the given event.

e.g.; ① Rolling a die

$$S = \{1, 2, 3, 4, 5, 6\}$$

$\therefore n = 6$ & Exhaustive cases
 $\downarrow$
Total cases

Let, $A = \text{getting } 1 \Rightarrow \{1\}$

Favourable cases, $m = 1$

$B = \text{getting } 2 \Rightarrow \{2\}$

$$m = 1$$

$C = \text{getting odd no.} \Rightarrow \{1, 3, 5\}$

$$m = 3$$

$D = \text{getting even no.} \Rightarrow \{2, 4, 6\}$

$$m = 3$$

e.g.; ② Tossing a coin twice.

$$S = \{HH, HT, TH, TT\}$$

$$\therefore n = 4$$

Let, A = getting two heads $\Rightarrow \{HH\}$

$$\therefore m = 1$$

B = getting 2 tails $\Rightarrow \{TT\}$

$$\therefore m = 1$$

C = getting 1 head $\Rightarrow \{HT, TH\}$

$$\therefore m = 2$$

D = getting at least 1 head $\Rightarrow \{HH, HT, TH\}$

$$\therefore m = 3$$

E = getting same face $\Rightarrow \{HH, TT\}$

$$\therefore m = 2$$

F = getting no head $\Rightarrow \{TT\}$

$$\therefore m = 1$$

G = -

③ Rolling two dice simultaneously.

$$S = \{11, 22, 33, 44, 55, 66, 12, 13, 14, 15, 16, 23, 24, 25, 26, 31, 32, 33, 34, 35, 36, 41, 42, 43, 44, 45, 46, 51, 52, 53, 54, 55, 56, 61, 62, 63, 64, 65, 66\}$$

$$S = \{11, 12, 13, 14, 15, 16, 21, 22, 23, 24, 25, 26, 31, 32, 33, 34, 35, 36, 41, 42, 43, 44, 45, 46, 51, 52, 53, 54, 55, 56, 61, 62, 63, 64, 65, 66\}$$

$$\therefore n = 36$$

Let, A = sum of 7

$\Rightarrow \{16, 25, 34, 43, 52, 61\}$

$\therefore n = 6$

B = sum less than 5

$\Rightarrow \{11, 12, 13, 21, 22, 31, 42\}$

$\therefore n = 6$

C = same face

$\Rightarrow \{11, 22, 33, 44, 55, 66\}$

$\therefore n = 6$

D = diff. face $\Rightarrow n = 30$

~~$\Rightarrow$~~

E = even no. in 2nd die

$\Rightarrow \{12, 14, 16, 22, 24, 26, 32, 34, 36, 42, 44, 46, 52, 54, 56, 62, 64, 66\}$

$\therefore n = 18$

(viii) Independent Events:

$\Rightarrow$ A and B are independent events

if $P(A) = P(A|B)$

or $P(B) = P(B|A)$

$\Rightarrow$ Two or more events are said to be independent if the occurrence of one event

does not affect the occurrence of another event.

(ix) Dependent events.

A and B are dependent events if

$$P(A) \neq P(A/B)$$

$$\text{or } P(B) \neq P(B/A)$$

Probability

Various Approaches (Definitions) to Probability,

(1) Classical or Mathematical Approach.

(2) Empirical or Statistical Approach

(3) Subjective Approach

(4) Axiomatic or Modern Approach

(1) Classical or ~~Mathematical~~ Approach: $\Rightarrow$ Equally likely events
 $\Rightarrow$ Mutually exclusive events

Consider a random experiment and there are 'n' no. of ~~same~~ exhaustive ~~case~~ mutually exclusive and equally likely outcomes (or cases or events). Out of n, m no. of outcomes are favorable to an event A (say).

Then, the probability of occurrence of event A is denoted by $P(A)$ and is given

$$\text{or } P(A) = \frac{\text{Favorable cases to event A}}{\text{Total Cases / Exhaustive Case}} = \frac{m}{n}$$

Similarly, the probability of non-occurrence of event A (i.e., occurrence of ~~comp~~ $\bar{A}$) is denoted by $P(\bar{A})$ and is given as:

$$P(\bar{A}) = \frac{\text{Favourable cases to } \bar{A}}{\text{Exhaustive cases}}$$

$$= \frac{n-m}{n}$$

$$= 1 - \frac{m}{n}$$

$$P(\bar{A}) = 1 - P(A)$$

This implies, $P(A) + P(\bar{A}) = 1$, i.e., total

probability = 1.

Eg: ① If a die is thrown, find the probability of getting,

i) odd no.

ii) multiple of 2

or Random experiment: a die is thrown Single stage random experiment

Sample Space (S) = {1, 2, 3, 4, 5, 6}

$\therefore$ Exhaustive cases, $n = 6$

② Let, $A =$ an odd no. $\rightarrow$ {1, 3, 5}

Favourable cases, $m = 3$

$$\therefore P(A) = \frac{m}{n} = \frac{3}{6} = \frac{1}{2}$$

② Let, $B = \text{multiple of } 2 \Rightarrow \{2, 4, 6\}$

$$\therefore m = 3$$

$$\therefore P(B) = \frac{m}{n} = \frac{3}{6} = \frac{1}{2}$$

eg: 2: If two dice are thrown simultaneously, then find the probability that the faces turned up show;

(i) sum less than 5

(ii) sum of 7

(iii) same faces

(iv) Different faces

Random Experiment:

Two dice are thrown

Multiple-stage random experiment

simultaneously

Sample Space : $(1,1), (1,2), (1,3), (1,4), (1,5),$

$(1,6), (2,1), (2,2), (2,3), (2,4),$

$(2,5), (2,6), (3,1), (3,2), (3,3),$

$(3,4), (3,5), (3,6) \dots (6,1), (6,2),$

$(6,3), (6,4), (6,5), (6,6)$

$\therefore$ Exhaustive cases, $n = 36$

(i) Favourable cases (m) = $\{(1,1), (1,2), (1,3), (2,1), (2,2), (3,1)\}$

$$= 6$$

$$\therefore P(\text{getting sum less than 5}) = \frac{m}{n} = \frac{6}{36} = \frac{1}{6}$$

(i) Let, $B = \text{sum of } \rightarrow \Rightarrow \{ (1,6), (2,5), (3,4), (4,3), (5,2), (6,1) \}$

$$\therefore n = 6$$

$$\therefore P(B) = 1/6$$

(ii) Let, $C = \text{sum faces} = 6 = n$

$$\therefore P(C) = 1/6$$

(iii) Different faces, $= D = 36 - 6 = 30$

$$\therefore P(D) = \frac{\cancel{36}^{15}}{\cancel{36}_{18}} = \frac{\cancel{18}^5}{\cancel{18}_6} = 5/6$$

Counting Rules

Fundamental Rules of Counting:

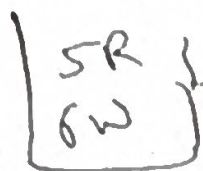
① Addition Rule

② Multiplication Rule

Addition Rule:

Consider a random experiment in which event A occurs in n_1 ways and another event B occurs in n_2 ways and are mutually exclusive events. Then, either A or B (at least one event) can occur in $(n_1 + n_2)$ ways.

eg;



A marble
is selected

A = Getting Red

B = Getting White

Here, A or B can occur in $5+6=11$ ways.

Multiplication Rule:

Consider a random experiment in which event A occurs in n_1 ways and event B occurs in n_2 ways. Then both A and B ~~can~~ can occur in $(n_1 \cdot n_2)$ ways.

eg;



Two marbles are
selected

A = Getting 1 Red

B = Getting 1 White

Here, A and B can occur in $5 \times 6 = 30$ ways.

Methods of Counting rules

① Permutation

② Combination

Permutation:

Arrangement of the objects.

A, B, C

Two objects are selected.

permutation
 $A B \neq B A$

It is a selection with order

Permutation of n distinct objects taken r at a time, is given by;

$${}^n P_r = \frac{n!}{(n-r)!} ; r \leq n$$

Note: ① $n! = n \cdot (n-1) \cdot (n-2) \cdots 3 \cdot 2 \cdot 1$

② $n! = n \cdot (n-1)!$

③ $0! = 1$

For repeated cases:

$$\text{Permutation} = \frac{n!}{a! b! c! \cdots}$$

Here, $n =$ Total no. of objects

$a =$ No. of repetition of object 1

$b =$ " " " " " 2

$c =$ " " " " " 3

Example: In how many ways the word
"STATISTICS" can be arranged?

Here;

Total = 10

$S = 3$

$T = 3$

$I = 2$

$A = 2$

$C = 1$

$$\therefore \frac{10!}{3! 3! 2! 2! 1!} = 1680$$

Combination

⇒ Selection of the objects without order.

A, B, C

↓ 2 objects are selected

A B = B A

↓
order doesn't matter. it's the same

Combination

Combination of n objects taken r at a time, is denoted by nC_r or $C(n, r)$ or $\binom{n}{r}$ and is given as:

$${}^nC_r = \frac{n!}{(n-r)! r!}; r \leq n$$

Q) How many sub-groups ~~is formed~~ of 4 persons can be formed, from a group of 3 scientists, 4 engineers and 5 economists?

Note,

$$n = 3 + 4 + 5 = 12$$

$$r = 4$$

$$\therefore {}^nC_r = \frac{12!}{(12-4)! 4!}$$

$$\begin{aligned} &= \frac{12 \times 11 \times 10 \times 9 \times \cancel{8!}}{\cancel{8!} \cdot 4! \cdot 5} \\ &= \frac{12 \times 11 \times 10 \times 9}{2 \times 2 \times 2 \times 2} = 495 \end{aligned}$$

Ex: From a group of 3 scientists, 4 engineers, and 5 economists, a sub-committee of 4 persons is selected at random. Find the probability that the sub-committee consists:

- Events
- ① 2 scientists, 1 engineer and 1 economist
 - ② 2 engineers and 2 economists
 - ③ All economists
 - ④ At least 1 scientist
 - ⑤ At most 2 engineers

Here,

Random Experiment:

Sub-committee of 4 persons is selected from 3 scientists, 4 engineers, and 5 economists. / ~~or~~ from 12 persons.

$\therefore$ Exhaustive cases $(n) = {}^{12}C_4 = 495$

① Let, $A =$ $\overset{\substack{\rightarrow 2 \text{ from } 3 \\ \text{scientists}}}{2 \text{ scientists}} \& \overset{\substack{\rightarrow 2 \text{ from } 4 \\ \text{engineers}}}{1 \text{ engineer}}$, and $\overset{\substack{\rightarrow 1 \text{ from } 5 \\ \text{economists}}}{1 \text{ economist}}$

$$\begin{aligned} \text{favourable cases, } m &= {}^3C_2 \times {}^4C_2 \times {}^5C_1 \\ &= 3 \times 4 \times 5 \\ &= 60 \end{aligned}$$

$$P(A) = \frac{m}{n} = \frac{60}{495}$$

② B = 2 engineers and 2 economists

favourable case, $m = {}^4C_2 \times {}^5C_2$

$$= \frac{4!}{2! 2!} \times \frac{5!}{3! 2!}$$
$$= 6 \times \frac{120}{6 \times 2}$$
$$= 60$$

$$\therefore P(A) = \frac{60}{495}$$

③ C = All economists

$\therefore$ favourable cases, $m = 5$

$$\therefore P(C) = \frac{5}{53} = \frac{5}{495}$$

2nd & 2nd 4
3rd

(4) At least 1 scientific = D $\Rightarrow$ 1 sci or more sci.

~~$$\therefore \text{Favourable cases, } n = {}^4C_1 + {}^3C_2$$~~

1 su or 2 or 3

2 su. 3 Ns.

12
v.

Non
Sien

$$\text{Favourable outcomes, } m = {}^3C_1 \times {}^9C_2 + {}^3C_2 \times {}^9C_1 + {}^3C_3 \times {}^9C_1$$

$$P(D) = \frac{3}{51} \approx 0$$

(5) Let, $E =$ At most 2 engineers

~~1 eng.~~
0 eng. & 4 NE or 1 eng. 3 NE
or 2 eng. 2 NE

$$= {}^9C_4 + {}^{12-4}C_4 + {}^4C_1 \times {}^8C_3 + {}^4C_2 \times {}^8C_2 \quad \#$$

Example:

(1) 2 bulbs are chosen at random from a lot containing 12 bulbs, of which 4 are defectives. What is the probability that;

- (i) both are defectives
- (ii) both are non-defectives
- (iii) one defective and one non-defective.

Soln

Total bulbs = 12

No. of defectives = 4

No. of non-defectives = 8

Random experiment:

2 bulbs are selected from 12 bulbs.

$\therefore$ Exhaustive cases $(n) = {}^{12}C_2 = 66$

(i) Let, A = Both are defective;

Favourable cases $m = {}^4C_2$

$$\therefore P(A) = \frac{m}{n} = \frac{6}{66} = \frac{1}{11}$$

(ii) Let, B = both are non-defective,

$$m = {}^8C_2 = 28$$

$$\therefore P(B) = \frac{28}{66} = \frac{14}{33}$$

(iii) Let, C = 1 defective & 1 non-defective

$$\therefore m = {}^4C_1 \times {}^8C_1 = 32$$

$$\therefore P(C) = \frac{32}{66} = \frac{16}{33}$$

Statistical or Empirical Approach:

This approach is used when the events are not equally likely.

Consider a random experiment and repeats for 'n' no. of times under similar condition.

Let A be an event that occurs 'm' times out of 'n' no. of repetition of a random experiment.

In limiting case when n becomes sufficiently large, then the probability of occurrence of event A i.e., $P(A)$ is finite and unique.

That is;

$$P(A) = \lim_{n \rightarrow \infty} \frac{m}{n}$$

Subjective Approach:

In this approach, the probability of occurrence of an event is defined on the basis of personal beliefs, knowledge, experience, intuition, opinions. This is biased approach since different individuals have different beliefs, knowledge, experiences, etc.

Axiomatic

Axiomatic or Modern Approach:

Let S be a sample space of a random experiment, and A be any event defined on sample space (S). Then, $P(A)$ is the probability function.

which satisfies the following axioms:

- i) $P(A) \geq 0 \Rightarrow$ Axiom of Non-Negativity
- ii) $P(S) = 1 \Rightarrow$ Axiom of certainty (or totality)

- iii) Let, $A_1, A_2, \dots, A_n$ be a finite or infinite ~~sequence~~ sequence of

mutually exclusive events. Then,

From the axiom of additivity, we have;

$$P\left(\bigcup_{i=1}^n A_i\right) = \sum_{i=1}^n P(A_i)$$

$\Downarrow$

$$P(A_1 \cup A_2 \cup \dots \cup A_n)$$

Also,

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i)$$

5th Laws of Probability: (Formula)

① Additive law of Probability

② Multiplicative law of Probability

③ Additive Law:

④ When two events A and B are not mutually exclusive events.

Then,

$$P(A \cup B) = P(A) + P(B) - \frac{P(A \cap B)}{A}$$

$\Downarrow$

Probability of occurrence of at least one event

(i.e., A or B)

Probability of occurrence of both events
(i.e., A and B)

Similarly, for three not mutually exclusive events, A, B, and C.

Then,

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C)$$

(ii) When two events A and B are mutually exclusive.

Then, $P(A \cup B) = P(A) + P(B)$

$$[\because P(A \cap B) = 0]$$

Similarly, for three mutually exclusive events A, B, and C.

Then, $P(A \cup B \cup C) = P(A) + P(B) + P(C)$.

Notes:

(1) If A be an event then the complementary event of A is denoted by $\bar{A}$ or A^c and the probability of complementary event $\bar{A}$ is given

as:

$$P(\bar{A}) = 1 - P(A)$$

$$\Rightarrow P(A) = 1 - P(\bar{A})$$

$$(2) P(\overline{A \cup B}) = 1 - P(A \cup B)$$

$$\Rightarrow P(A \cup B) = 1 - P(\overline{A \cup B})$$

$\Downarrow$

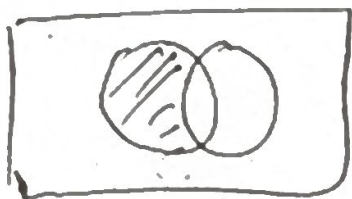
$$P(A \cap \bar{B})$$

③ For any two events A and B. Then,

$$P(A \cap \bar{B}) = P(A) - P(A \cap B)$$



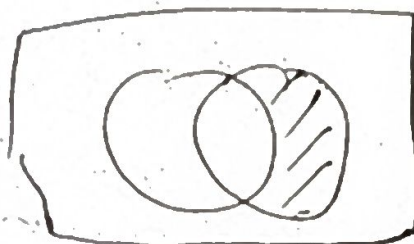
Prob
P(A only)



$$P(\bar{A} \cap B) = P(B) - P(A \cap B)$$



P(B only)



Multiplicative law:

① When two events ~~are independent~~
A and B are independent. Then,

$$P(A \cap B) = P(A) \times P(B)$$

Similarly, for three independent events

A, B and C. Then,

$$P(A \cap B \cap C) = P(A) \cdot P(B) \cdot P(C)$$

② when two events A and B are dependent events. Then,

$$P(A \cap B) = \begin{cases} P(A) \cdot P(B|A) \\ P(B) \cdot P(A|B) \end{cases}$$

where, $P(A|B)$ = Conditional probability of A given that B has already occurred

& $P(B|A)$ = Conditional Probability of B given that A has already been occurred.

Note:

If A and B are two independent events. then,

- (i) $\bar{A}$ and $\bar{B}$ are also independent
- (ii) A and $\bar{B}$ are also " "
- (iii) $\bar{A}$ and B are " "

Also, $P(A \cap \bar{B}) = P(A) \cdot P(\bar{B})$

and, $P(\bar{A} \cap B) = P(\bar{A}) \cdot P(B)$

$$\begin{aligned} P(A \cap \bar{B}) &= P(A) - P(A \cap B) \\ &= P(A) - P(A) \cdot P(B) \\ &= P(A) \{1 - P(B)\} \\ &= P(A) \cdot P(\bar{B}) \end{aligned}$$

9) In a city a survey of 600 families about the fuel for cooking reveal that 230 use LP gas, 170 use induction and 40 use both. then find the probability that a randomly selected family use,

- (i) at least one of the fuel
- (ii) neither of the fuel
- (iii) exactly one type of fuel.

Given

Total families = 600 $\rightarrow n(S)$

No. of families use LPG = 230 $\rightarrow n(L)$

No. of families use Induction = 175 $\rightarrow n(I)$

No. of families use both = 40 $\rightarrow n(L \cap I)$

Define the events:

A = family use LPG

B = family use Induction

Then, C_0

$$P(A) = \frac{230}{600} = \frac{23}{60}$$

$$P(B) = \frac{175}{600} = \frac{35}{120} = \frac{7}{24}$$

$$P(A \cap B) = \frac{40}{600} = \frac{2}{30} = \frac{1}{15}$$

(i) At least one fuel;
A or B

$$P(A \text{ or } B) = P(A \cup B)$$

$$= P(A) + P(B) - P(A \cap B)$$

$$= \frac{73}{120}$$

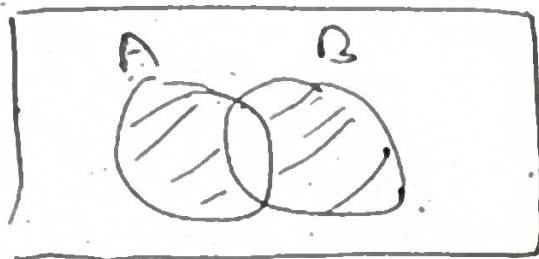
(ii) Neither of fuel

not A and B $\Rightarrow A \cap B \Rightarrow \overline{A \cup B}$

$$P(\overline{A \cup B}) = 1 - P(A \cup B)$$

iii) exactly one type of fuel.

$$A \text{ only or } B \text{ only} = \frac{P(A \cup B)}{P(A \cap B)} \rightarrow A_0 \text{ or } B_0$$



$$P(A_0 \cup B_0) = [P(A_0) + P(B_0)] - P(A_0 \cap B_0)$$

$$= P(A \cap B) + P(\bar{A} \cap B) + [P(B) - P(A \cap B)]$$

8) A problem of statistics is given to 3 students A, B, and C whose chances of solving it are $\frac{1}{2}$, $\frac{1}{3}$, and $\frac{1}{4}$ respectively. Find the probability that

- (i) All of them can solve the problem
- (ii) None of them can solve
- (iii) Problem will be solved (at least one can solve the problem).
- (iv) A can solve but B and C cannot solve.
- (v) A and B can solve but C cannot solve.
- (vi) Only one of them can solve.
- (vii) Only two of them can solve.

iii

Define the events:

$A = A$ solves the problem

$B = B$ solves the problem

$C = C$ solves the problem

Here, $A, B,$ and C are independent events

Then, $P(A) = \frac{1}{2}, P(B) = \frac{1}{3}, P(C) = \frac{1}{4}$

$\bar{A} = A$ doesn't solve the problem

$\bar{B} = B$ " " " "

$\bar{C} = C$ " " " "

$$P(\bar{A}) = 1 - P(A) = \frac{1}{2}$$

$$P(\bar{B}) = 1 - \frac{1}{3} = \frac{2}{3}$$

$$P(\bar{C}) = 1 - \frac{1}{4} = \frac{3}{4}$$

i) All of them can solve:

A solves and B solves and C solves

$$= \cancel{P(A) \cdot P(B) \cdot P(C)} \quad P(A \cap B \cap C)$$

$$= \cancel{\frac{1}{2} \cdot \frac{1}{3} \cdot \frac{1}{4}} \quad P(A) \cdot P(B) \cdot P(C)$$

$$= \frac{1}{2} \cdot \frac{1}{3} \cdot \frac{1}{4}$$

$$= \frac{1}{24}$$

ii) None of them can solve:

A doesn't solve and B doesn't solve and C

doesn't solve

$$= P(\bar{A} \cap \bar{B} \cap \bar{C})$$

$$= P(\bar{A}) \cdot P(\bar{B}) \cdot P(\bar{C})$$

$$= \frac{1}{2} + \frac{2}{2} + \frac{3}{4}$$

$$= \frac{1}{4}$$

iii) Problem will be solved;

at least one of them can solve.
 either
 $\Rightarrow A$ solves or B solves or C solves

$$= P(A \cup B \cup C)$$

$$= P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C)$$

$$= \frac{1}{2} + \frac{1}{3} + \frac{1}{4}$$

$$= \frac{12+8+6}{24}$$

$$= \frac{26}{24}$$

$$= \frac{13}{12}$$

(iv) A can solve but B and C can't;
 $\Rightarrow$ A solves and B doesn't solve

$$= P(A \cup B \cup C)$$

$$= 1 - P(\bar{A} \cap \bar{B} \cap \bar{C})$$

$$= 1 - P(\bar{A} \cap \bar{B} \cap \bar{C})$$

$$= 1 - \frac{1}{4}$$

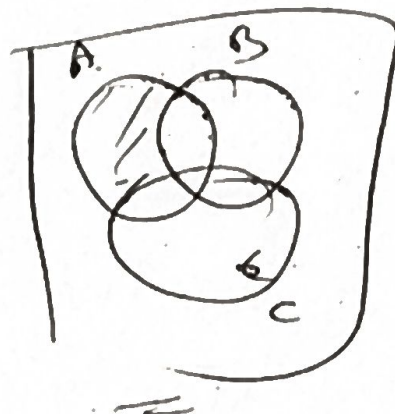
$$= \frac{3}{4}$$

∴ A can solve but B and C can't.

⇒ A solves and B doesn't solve and C doesn't solve

$$= P(A \cap \bar{B} \cap \bar{C})$$
$$= P(A) \cdot P(\bar{B}) \cdot P(\bar{C})$$

$$= \frac{1}{2} \cdot \frac{2}{3} \cdot \frac{3}{4} = \frac{1}{4}$$



$P(A_{\text{only}})$

③

$$= 1 - P(B \cup C) \checkmark$$

∴ A & B can but C cannot:

$$= P(A \cap B \cap \bar{C})$$

$$= P(A) \cdot P(B) \cdot P(\bar{C})$$

$$= \frac{1}{2} \cdot \frac{1}{3} \cdot \frac{3}{4}$$

$$= \frac{1}{8}$$

vi) only one of them can solve:

↳ either A or B or C only A or only B or only C solve

$$P(A \cup B \cup C)$$

$$= P(A \cup B \cup C)$$

$$= P(A) + P(B) + P(C)$$

$$= P(A) + P(B) + P(C)$$

$$\left\{ \begin{aligned} P(A_{\text{only}}) &= P(A \cap \bar{B} \cap \bar{C}) \\ &= P(A) - P(A \cap C) - P(A \cap B) + P(A \cap B \cap C) \end{aligned} \right\}$$

$\rightarrow P(A_{\text{only}}) = P(A) - P(A \cap B) - P(A \cap C) + P(A \cap B \cap C)$

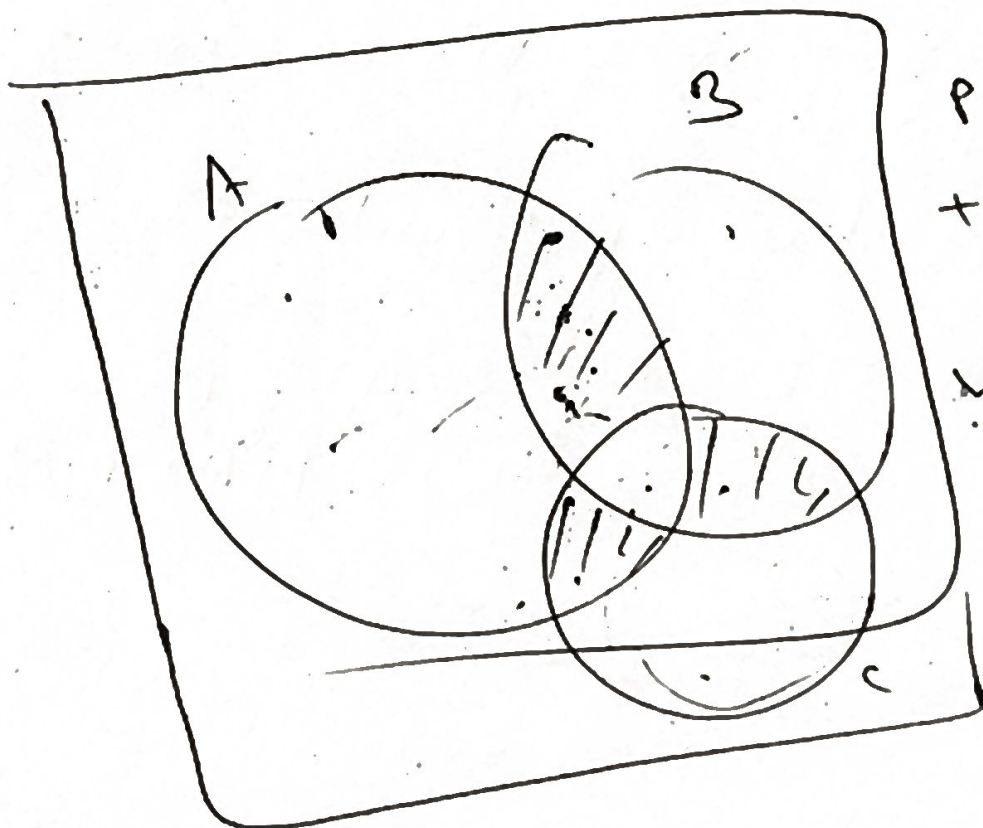
$$\begin{aligned} &= \{ P(A \cap \bar{B} \cap \bar{C}) + P(\bar{A} \cap B \cap \bar{C}) + P(\bar{A} \cap \bar{B} \cap C) \\ &= \{ P(A) \cdot P(\bar{B}) \cdot P(\bar{C}) + P(\bar{A}) \cdot P(B) \cdot P(\bar{C}) + \\ &\quad P(\bar{A}) \cdot P(\bar{B}) \cdot P(C) \} \end{aligned}$$

=

vii) Only two of them:

$\Rightarrow$ (A and B) only but not C or (B and C) only but not A
or (A and C) only but not B

$$= P(A \cap B)$$



$$\begin{aligned} &P(A \cap B \cap \bar{C}) \\ &+ P(B \cap C \cap \bar{A}) \\ &+ P(A \cap C \cap \bar{B}) \end{aligned}$$

$$\begin{aligned} &= [P(A \cap B) - P(A \cap B \cap C)] + [P(A \cap C) - P(A \cap B \cap C)] \\ &\quad + [P(B \cap C) - P(A \cap B \cap C)] \\ &= P(A) \cdot P(B) + P(A) \cdot P(C) + P(B) \cdot P(C) - 3 \cdot P(A \cap B \cap C) \end{aligned}$$

Q) The chance of winning the race of two horses are $\frac{1}{3}$ and $\frac{1}{6}$ respectively. Find the probability that at least one horse will win the race if they are running;

(i) In the different races

(ii) In the same race

→ Define the events:

(i) A = horse 1 win the race

B = horse 2 win the race

Then, $P(A) = \frac{1}{3}$, $P(B) = \frac{1}{6}$

→ In the different races;

Here, A and B are not mutually exclusive (in different races both can win) and independent events.

Now, at least one will win the race?

$$\begin{aligned} P(A \cup B) &= P(A) + P(B) - P(A \cap B) \\ &= P(A) + P(B) - P(A) \cdot P(B) \\ &= \end{aligned}$$

(ii) In the same race;

Here, A & B are mutually exclusive.

$$P(A \cup B) = P(A) + P(B)$$

at least one can win

Eg; The odds against A solving the problem is $8:6$ and odds in favour of B ~~and~~ solving the same problem is $3/4:10$. What is the probability that,

- (i) Both A and B can solve the problem
- (ii) A solves but B fails to solve
- (iii) At least one can solve.

Note: Odds $\rightarrow$ ratio

(1) Odds in favour of Event A
Ratio of Favourable cases of A to unfavourable cases of A.

Odds in favour of A is $m:n$

$\swarrow$ favourable cases of A $\searrow$ Unfavourable cases of A

Then,

$$P(A) = \frac{m}{m+n}$$

(2) Odds against event A
Ratio of unfavourable cases of A to favourable cases of A.

eg: odds against A is $m:n$

$\swarrow$ unfavourable cases $\searrow$ favourable cases

Then, $P(A) = \frac{n}{m+n}$

$$P(A) = \frac{6}{14} = \frac{3}{7}$$

$$P(B) = \frac{14}{24} = \frac{7}{12}$$

i) $P(A \cap B) = P(A) \cdot P(B)$

ii) $P(A \cap \bar{B}) = P(A) \cdot P(\bar{B})$ ~~$= P(A) \cdot P(\bar{B})$~~
 ~~$= P(A) \cdot (1 - P(B)) = P(A) \cdot P(\bar{B})$~~

iii) $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

Q.3 Three groups of children contain three girls, two boys; two girls, two boys; and 1 girl, and two boys respectively. One child is selected from each group, what is the probability that the selected children will be;

(i) 1 girl and two boys. $\rightarrow$ 1 child selected

G1	G2	G3 $\rightarrow$ 1 child is selected
3G & 2B	2G & 2B	1G & 2B

one child is selected from each grp.

1 child

is selected

interm.

(i) 1G & 2B

	G1	G2	G3
Events			
A	G	B	B
B	B	G	B
C	B	B	G

mutually exclusive

For event A:

$$P(A) = P(G_1 \text{ and } B_2 \text{ and } B_3)$$

$$= P(G_1) \cdot P(B_2) \cdot P(B_3) \quad [\because \text{they are independent}]$$

$$= \frac{3}{5} \cdot \frac{2}{4} \cdot \frac{2}{3}$$

$$= \frac{6}{30} = \frac{1}{5} \quad \text{---}$$

Similarly,

$$P(B) = P(B_1 \text{ and } G_2 \text{ and } B_3)$$

$$= P(B_1) \cdot P(G_2) \cdot P(B_3)$$

$$= \frac{2}{5} \cdot \frac{2}{4} \cdot \frac{2}{3}$$

$$= \frac{4}{30}$$

$$= \frac{2}{15}$$

$$\& P(C) = P(B_1 \& B_2 \& G_3)$$

$$= P(B_1) \cdot P(B_2) \cdot P(G_3)$$

$$= \frac{2}{5} \cdot \frac{2}{4} \cdot \frac{1}{3}$$

$$= \frac{2}{30}$$

$$= \frac{1}{15}$$

Now, at least one event can occur. A, B, C are mutually exclusive

$$P(A \cup B \cup C) = P(A) + P(B) + P(C)$$

$$= \frac{1}{5} + \frac{2}{15} + \frac{1}{15}$$

Q3 The odds that A speaks the truth are 3:2 and the odds that B speaks the truth are 5:3. What is the probability that they contradict each other on an identical point.

probability of A speaking truth;

$$P(A) = \frac{3}{5}$$

probability of B speaking truth;

$$P(B) = \frac{5}{8}$$

∴ probability of them contradicting each other.

$P(A \text{ speaks truth and } B \text{ doesn't}) \text{ or } (A \text{ ~~speaks~~ doesn't speak truth and } B \text{ speaks})$

$$= P(A \cap \bar{B}) + P(\bar{A} \cap B)$$

$$= P(A) \cdot P(\bar{B}) + P(\bar{A}) \cdot P(B)$$

$$= \frac{3}{5} \left\{ 1 - \frac{5}{8} \right\} + \frac{2}{5} \left\{ 1 - \frac{3}{5} \right\} \cdot \frac{5}{8}$$